Eva Liu

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EDUCATION

Georgia Institute of Technology, College of Computing Incoming Jan 2027
- Master of Science in Computer Science

University of Washington, Paul G. Allen School of Computer Science & Engineering Sep 2020 – Dec 2024
- Bachelor of Science in Computer Science

TECHNICAL SKILLS

- TypeScript (Next.js, React), Python, HTML, CSS, Supabase, Fast API, Voyage AI embeddings, pgvector, and semantic search

SOFTWARE EXPERIENCE

Z&V Home Construction LLC, Software Engineer Contract Jun 2026 – Present
- Rebuilt the company's Wix site as a custom responsive **Next.js** application in **React, TypeScript, and Tailwind CSS**, roughly tripling daily traffic within two weeks of launch and reaching an 11.6% search click-through rate, with a project gallery, service sections, and a lead capture contact form.
- Owned deployment and discoverability end to end, configuring hosting, a custom domain, HTTPS, and automatic builds from source control, then implementing page metadata, Open Graph tags, and a sitemap and creating the business's first Google Search Console property to get the site indexed and measurable for local search.
- Maintained the live site as the sole developer, handling content updates, image and performance optimization, dependency upgrades, and new feature requests.

Center for Reproducible Biomedical Modeling (CRBM), Software Engineer Jul 2022 – Jan 2025
- Led a team of student engineers building a browser-based computational biology modeling editor, letting researchers create models with zero local setup.
- Owned end-to-end execution across roadmap, sprint planning, and delivery, converting requirements into shipped features.
- Presented at **international standards conferences** (COMBINE, HARMONY) and **co-authored a peer-reviewed Oxford Bioinformatics paper** on the VSCode Antimony extension (DOI: 10.1093/bioinformatics/btad753).
- Built a React/TypeScript in-browser editor for the Antimony modeling language on Monaco Editor, implementing grammar recognition, syntax highlighting, and contextual hover documentation.
- Developed core features for the VSCode Antimony extension **(6,000+ Marketplace downloads)**, including variable annotations and dynamic rate law insertion via VS Code APIs, cutting manual modeling effort.
- Engineered an automated **Python** and TypeScript installation pipeline, removing setup friction for new users.

SIDE PROJECTS

Nodemic Epidemic Modeler Sep 2026 – Present
- Building a browser app where users draw disease-spread models as connected boxes (Susceptible, Infected, Recovered) and watch the outbreak play out on a live chart.
- Wrote the engine that turns the drawn diagram into math and steps it forward day by day, checked against textbook outbreak results.
- Debounced and memoized edits so the chart recalculates once after the user stops typing or dragging.

Blog Platform Oct 2025 – Sep 2026
- Built a full-stack blogging platform using **Next.js, React, TypeScript, Python, HTML, CSS and Supabase with Wisp CMS** for structured content management and dynamic rendering.
- Developed an **AI-powered article recommendation** system using **Voyage AI embeddings, pgvector, and semantic search** to surface related articles.
- Developed backend services with **FastAPI** to generate, store, and retrieve vector embeddings to improve user content discovery.
- Implemented modular frontend architecture to support reusable layouts and scalable content delivery.

Find Your Needs Mar 2024 – Jun 2024
- Built a cross-platform Flutter/Dart app (iOS and Android) surfacing food, hygiene, and library resources across greater Seattle, rendering a typed location dataset as icon-differentiated Google Maps markers alongside a distance-sorted list view with deep-link handoff to the device's native navigation app.
- Implemented real-time geolocation via a Provider state layer feeding live distance and marker updates, with session persistence for map position and zoom through shared preferences.